Modeling of the experimental part

3.1 Goal of the experiments

We want to see if our modified PID controller keeps the water level close to the set point and moves the valve less than a normal PID, when the inflow and outflow are unknown. We also want to know what each extra part of the controller does.

3.2 The simulated process

The process is a tank with a valve. We use a simple model of the water balance, and it is the model of the class TankSimulator.

The variables are:

- PV(k): the water level at step k, in percent of the tank (from 0 to 100).
- SP(k): the set point, the level we want.
- u(k): the valve position, from 0 to 100.
- u0(k): the null position of the valve, where the inflow is equal to the outflow, so the level does not change. The controller does not know it. In S1 it is 50 all the time, and in other situations it can change.
- d(k): a disturbance added to the level, like rain (positive) or evaporation (negative). It is 0 in all the situations, except S4.
- r(k): the gain of the valve at step k, a random number.